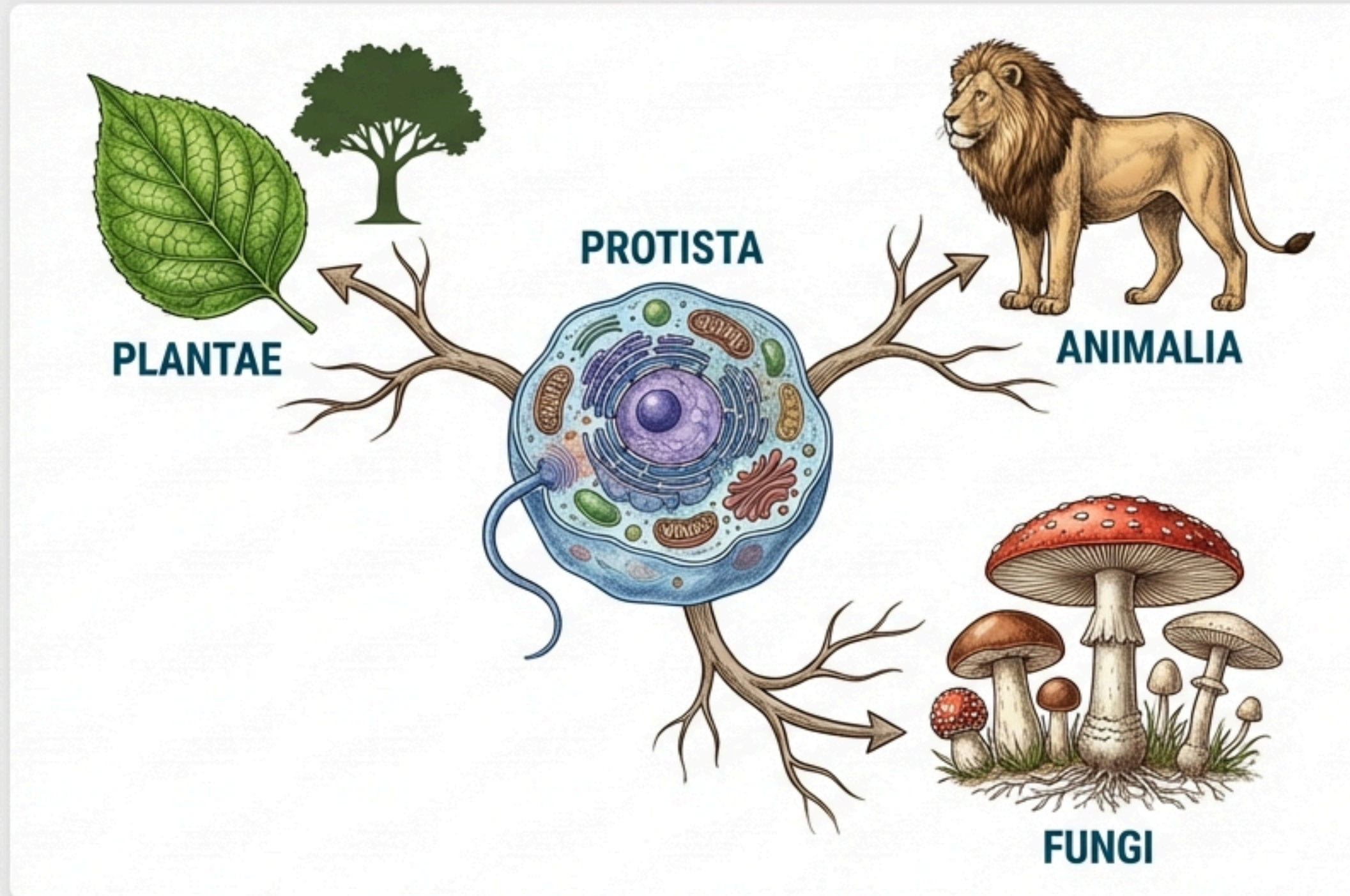


Kingdom Protista: The Evolutionary Link

From Undefined Boundaries to Eukaryotic Complexity



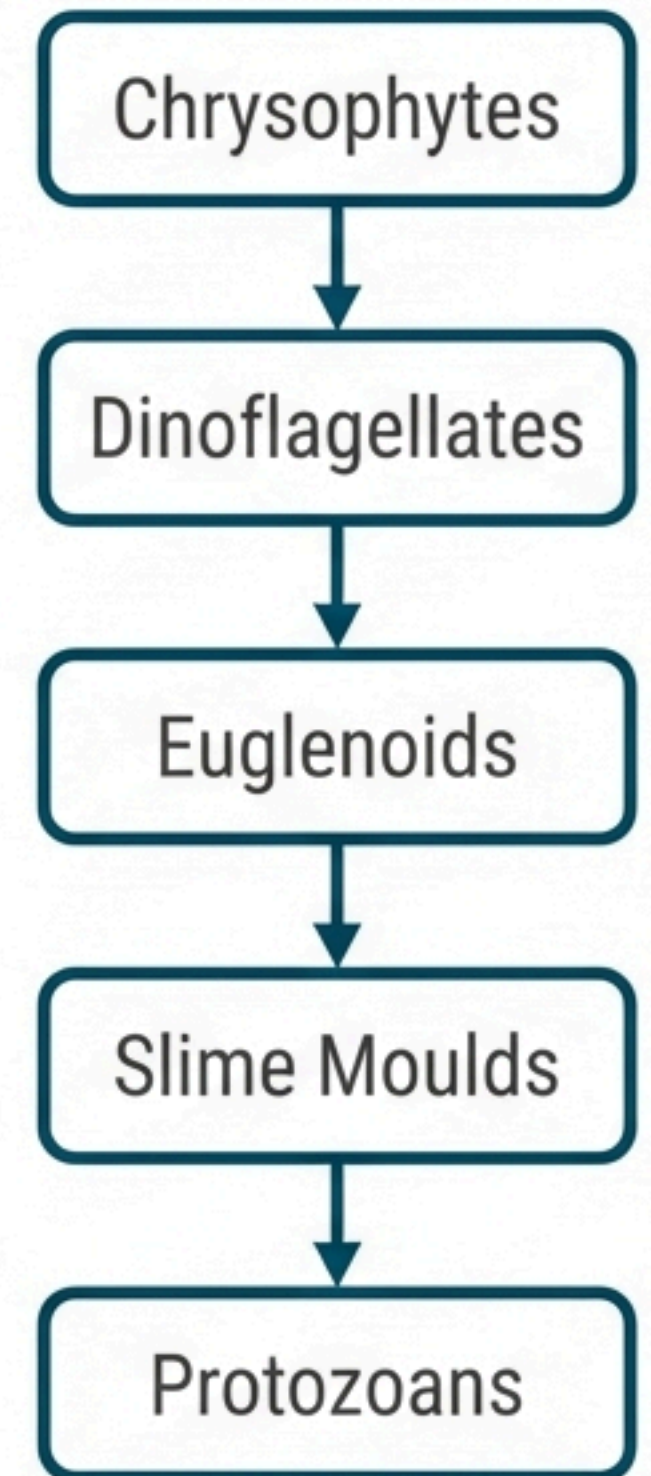
General Characteristics & Classification

All **single-celled eukaryotes** are placed under Protista, but the **boundaries of this kingdom are not well defined.**

EXAM ALERT: Potential Statement Question: “What may be a ‘photosynthetic protistan’ to one biologist may be a ‘plant’ to another.”

Key Features

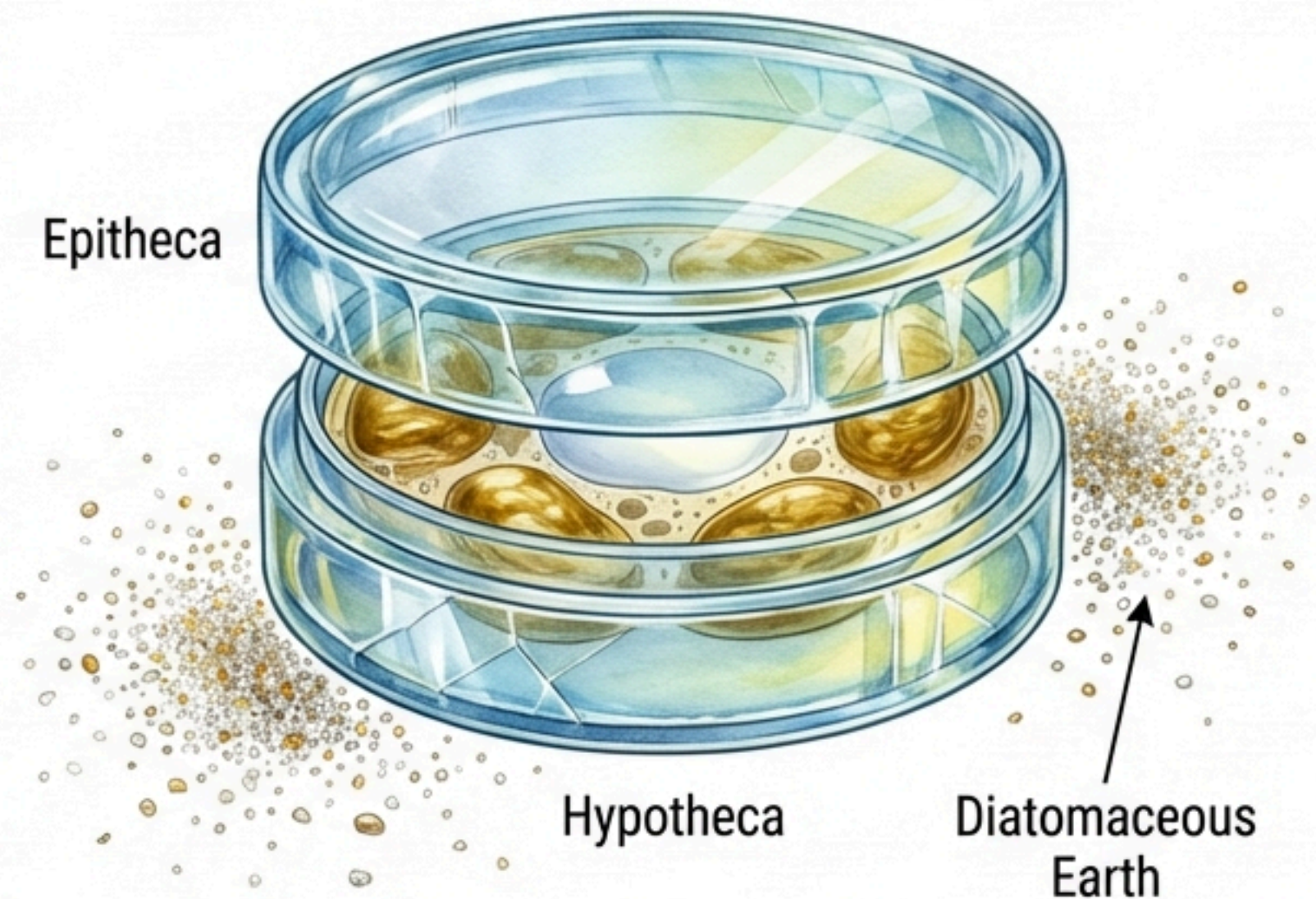
- **Habitat:** Primarily aquatic.
- **Cell Structure:** Well-defined nucleus and membrane-bound organelles.
- **Locomotion:** Some possess flagella or cilia.
- **Reproduction:** Asexual and Sexual (involving **cell fusion** and **zygote formation**).



Chrysophytes (Diatoms & Golden Algae)

- **Habitat:** Fresh water and marine environments.
- **Lifestyle:** Microscopic; float **passively** in water currents (**Plankton**). Most are photosynthetic.
- **Cell Walls:** Form two thin overlapping shells which fit together as in a **soap box**.
- **Composition:** Walls are embedded with **silica**, making them **indestructible**.
- **Ecological Impact: Diatomaceous Earth** (accumulation of wall deposits over billions of years). Used in polishing, filtration of oils, and syrups.

EXAM ALERT: Diatoms are the **chief 'producers'** in the oceans.



Dinoflagellates (The Armored Protists)

Habitat/Nutrition: Mostly marine and photosynthetic.

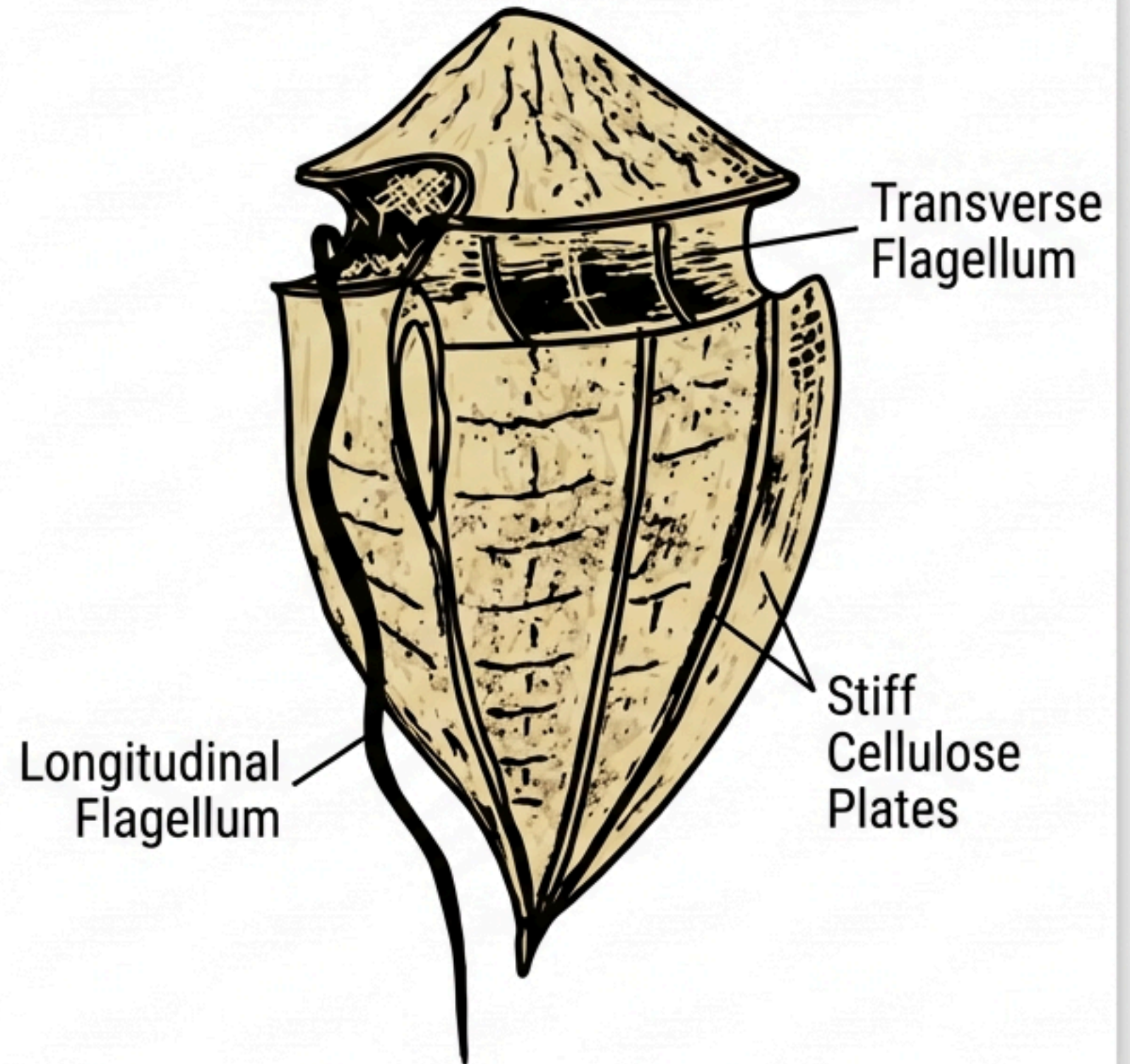
Appearance: Yellow, green, brown, blue, or red.

Cell Wall: Stiff cellulose plates on the outer surface.

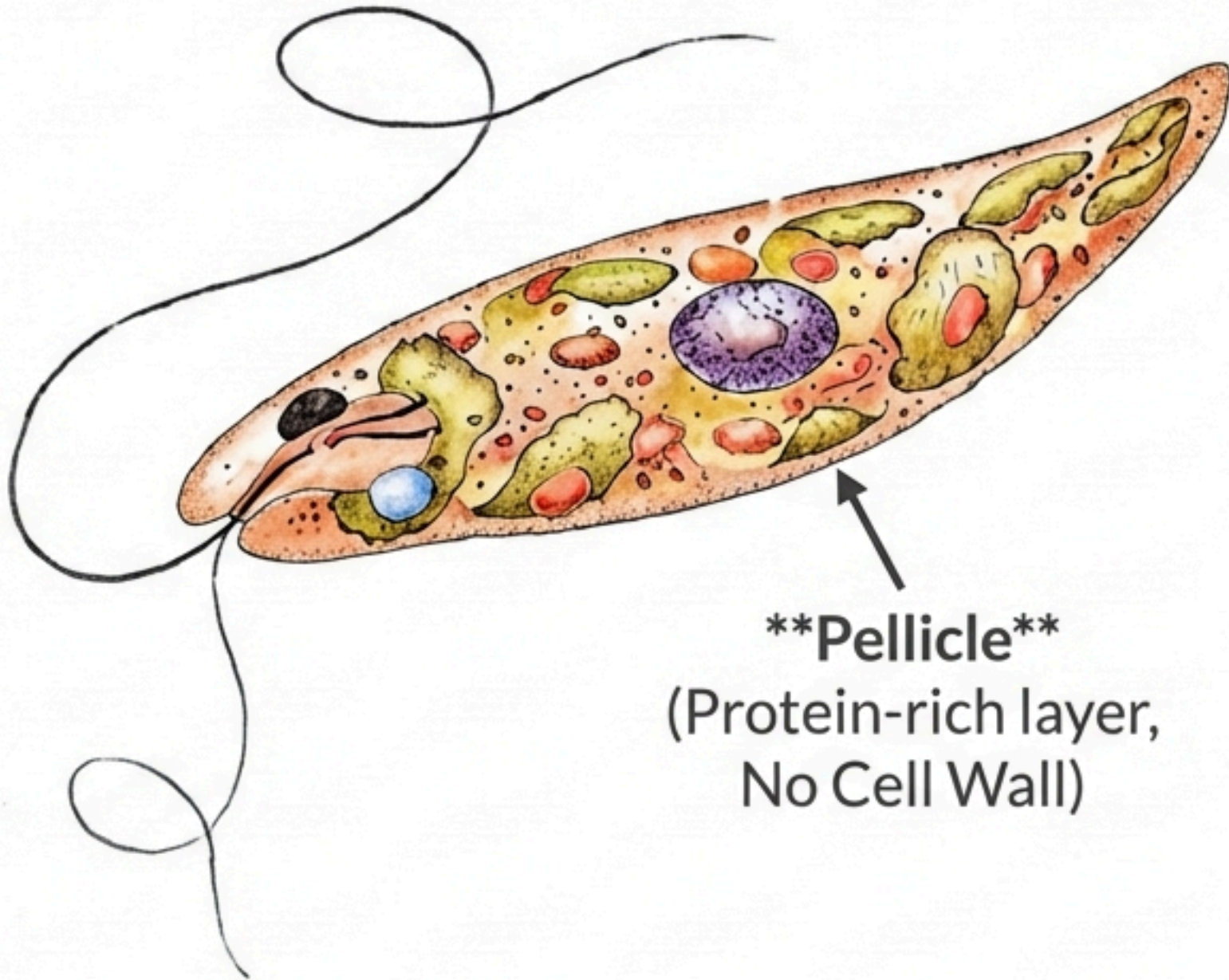
Flagellation: Two flagella (Longitudinal and Transverse in a furrow).

Red Tides: Caused by rapid multiplication of Red Dinoflagellates (Example: *Gonyaulax*).

Consequence: Toxins released can kill marine animals.



Euglenoids (The Flexible Hybrids)



****Pellicle****
(Protein-rich layer,
No Cell Wall)

Habitat: Fresh water (**stagnant water**).

Structure: No cell wall. Have a **pellicle** which makes the body flexible.

Flagella: Two (one short, one long).

Mixotrophic Nutrition:

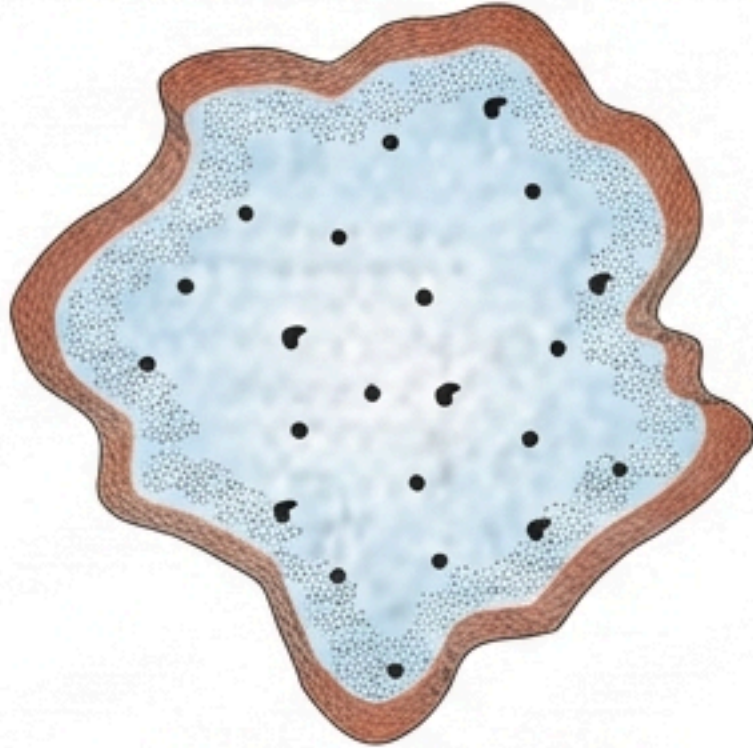
- Sunlight present: **Photosynthetic**.
- Sunlight absent: **Heterotrophs** (predators).

Exam Alert

The pigments of euglenoids are **identical** to those present in **higher plants**.

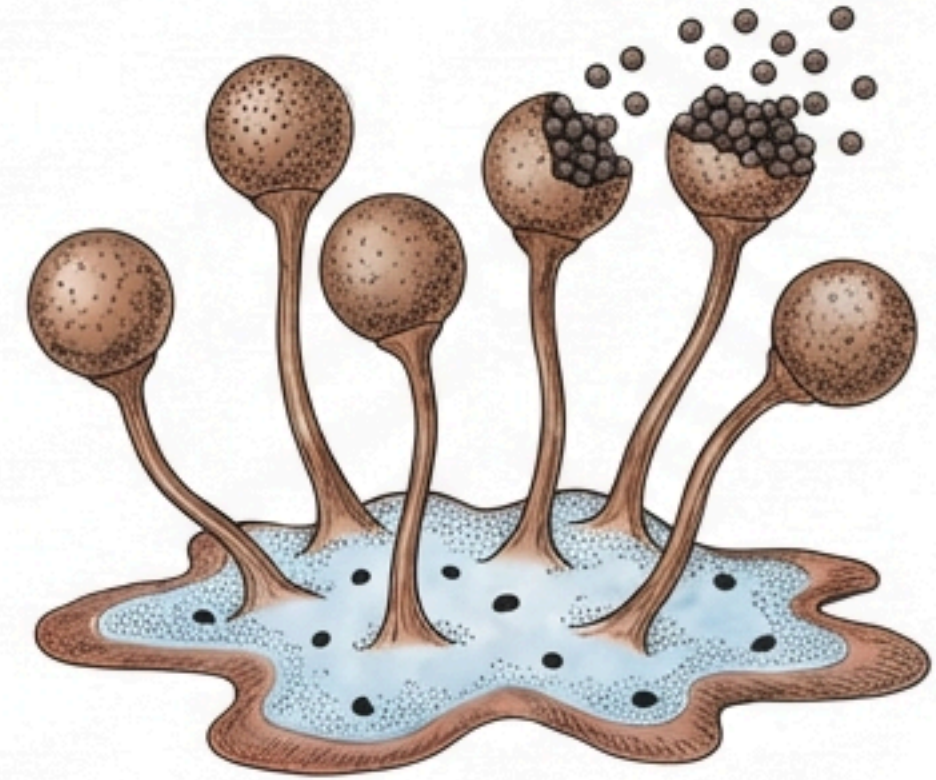
Slime Moulds (Saprophytic Protists)

Suitable Conditions



Plasmodium

Unfavorable
Conditions



Fruiting Bodies

Plasmodium

Aggregation formed under suitable conditions.
Grows and spreads over several feet.
Engulfs organic material.

Spores

Formed at tips of fruiting bodies.
Possess **true walls**.
Extremely resistant (survive many years).
Dispersed by **air currents**.

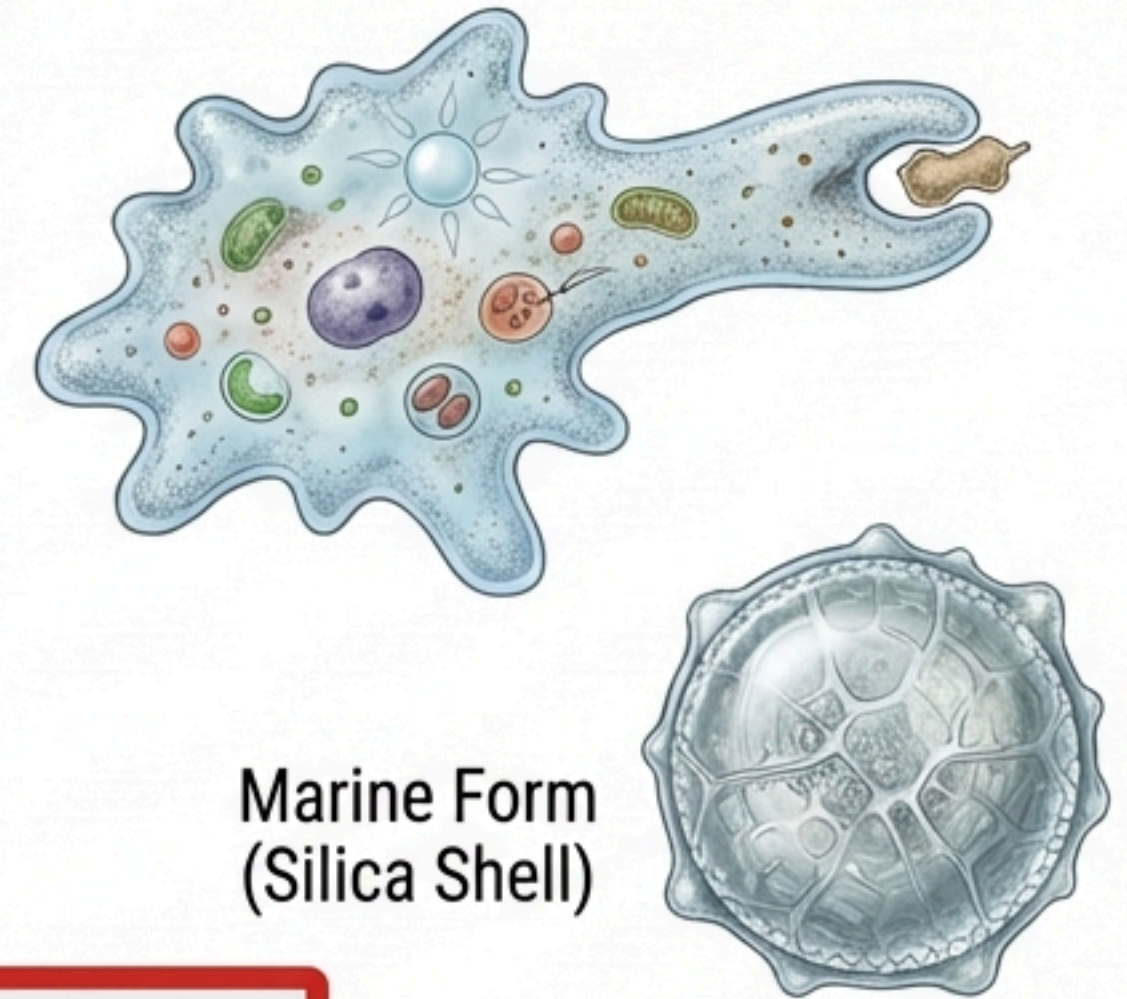
Exam Note:
Plasmodium = No wall mentioned;
Spores = True walls.

Protozoans: Primitive Relatives of Animals

All are heterotrophs (predators or parasites).

Group 1: Amoeboid Protozoans

- **Habitat:** Fresh water, sea water, or moist soil.
- **Locomotion:** Move and capture prey via **pseudopodia** (false feet). Example: *Amoeba*.
- **Marine forms:** Have **silica shells** on their surface.
- **Parasitic forms:** Example: *Entamoeba*.



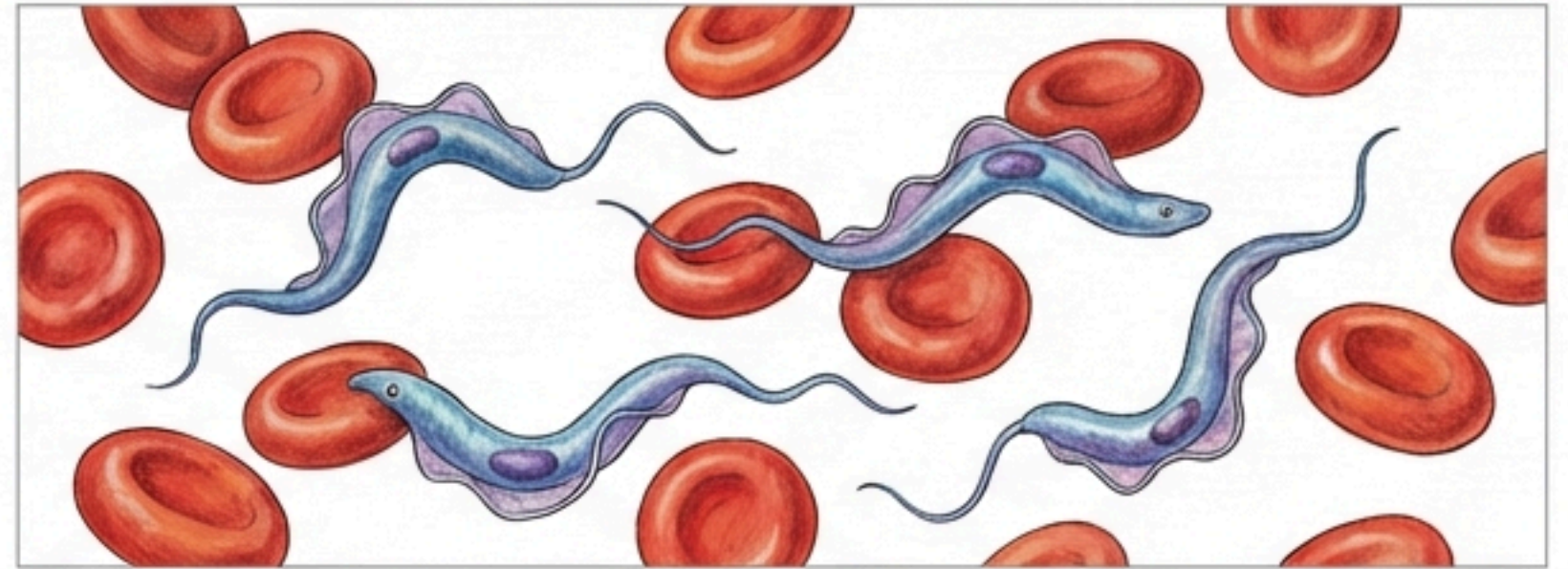
Exam Alert: Don't confuse Silica shells in Marine Amoeboids with Silica embedded walls in Chrysophytes.

Flagellated & Ciliated Protozoans

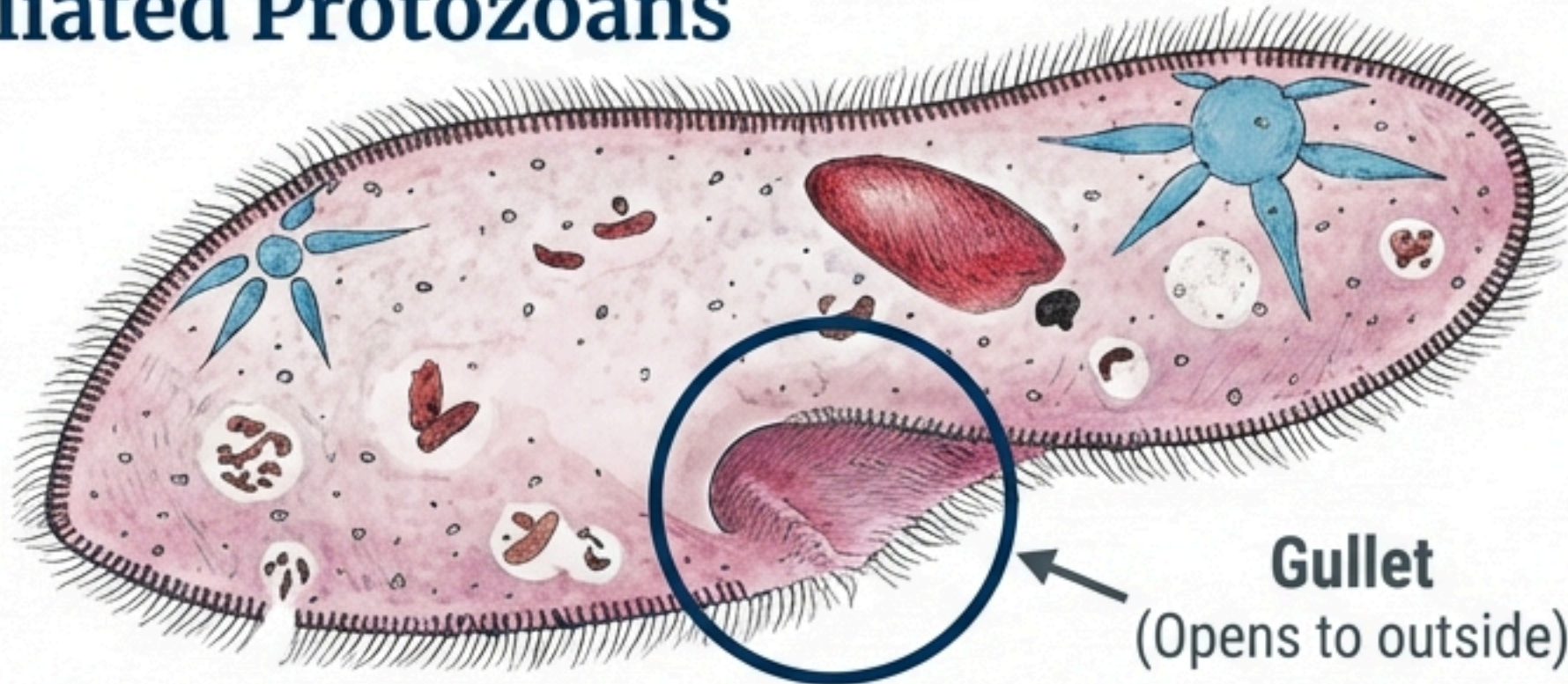
Flagellated Protozoans

Free-living or parasitic. Have flagella.

- Disease: **Sleeping Sickness** caused by *Trypanosoma*.



Ciliated Protozoans



- Actively moving due to thousands of **cilia**.
- Feeding: Coordinated movement of cilia steers water laden with food into the **gullet**.
- Example: *Paramecium*.

Sporozoans

Sporozoans: Key Characteristics & Impact

- **Key Characteristic:** Diverse organisms that have an **infectious spore-like stage** in their life cycle.
- **Notorious Example:** *Plasmodium* (Malarial parasite).
- **Impact:** Causes **malaria**, a disease with a staggering effect on human population.

Protozoan Quick-Summary Table

Group	Locomotion	Key Feature
Amoeboid	Pseudopodia	Silica shells (marine)
Flagellated	Flagella	Parasitic (Sleeping Sickness)
Ciliated	Cilia	Gullet mechanism
Sporozoans	None mentioned	Infectious spore-like stage

The High-Yield Comparison Matrix

Group	Cell Wall / Layer	Locomotion	Nutrition	Example
Chrysophytes	Silica embedded (Soap box)	Passive float (currents)	Photosynthetic	Diatoms
Dinoflagellates	Stiff Cellulose Plates	2 Flagella (Longit/Transv)	Photosynthetic	<i>Gonyaulax</i> (Red Tide)
Euglenoids	Protein-rich Pellicle	2 Flagella	Mixotrophic	<i>Euglena</i>
Slime Moulds	Spores have true walls	Amoeboid (<i>Plasmodium</i>)	Saprophytic	<i>Plasmodium</i>
Protozoans	None / Silica Shells (some)	Pseudopodia/ Cilia/Flagella	Heterotroph	<i>Amoeba</i> , <i>Paramecium</i>

Exam Drill: Statement Based Questions (Part 1)

Q1. Which of the following statements regarding Chrysophytes is INCORRECT?

- A) They are found in fresh water as well as in marine environments.
- B) Their cell walls are embedded with calcium carbonate making them hard.
- C) They float passively in water currents.
- D) Diatomaceous earth is used in filtration of oils.

Q2. Read the following statements about Dinoflagellates:

- I. Most are marine and photosynthetic.
- II. They have stiff cellulose plates on the inner surface of the cell wall.
- III. *Gonyaulax* is responsible for red tides.

Select the correct statements:

- (A) I only (B) I and III (C) II and III (D) All of the above

Q3. Identify the correct statement regarding Euglenoids:

- A) They possess a rigid cell wall made of chitin.
- B) Their pigments are identical to those present in primitive algae.
- C) In the absence of sunlight, they behave like heterotrophs.
- D) They are primarily marine organisms.

Exam Drill: Assertion & Reason (Part 2)

Directions: (A) Both True & R explains A, (B) Both True but R does not explain A, (C) A True / R False, (D) A False / R True

Q4:

Assertion (A): Slime moulds form fruiting bodies during unfavorable conditions.

Reason (R): The spores produced in fruiting bodies possess true walls and are extremely resistant.

Q5:

Assertion (A): Diatoms are considered the chief producers in the oceans.

Reason (R): Diatom cell walls are embedded with silica and are indestructible.

Q6:

Assertion (A): Ciliated protozoans are actively moving organisms.

Reason (R): They possess thousands of cilia and a gullet that opens to the outside.

Exam Drill: Match & Identify (Part 3)

Q7. Match the Columns:

- | | | |
|-------------------------|-------|-------------------------|
| (a) Saprophytic Protist | --- | (i) <i>Paramecium</i> |
| (b) Golden Algae | ----- | (ii) <i>Plasmodium</i> |
| (c) Sleeping Sickness | ----- | (iii) Desmids |
| (d) Gullet | ----- | (iv) <i>Trypanosoma</i> |

Q8. Which organism is known to have a protein-rich layer called pellicle?

- | | |
|----------------------|-----------------------|
| (A) <i>Gonyaulax</i> | (B) <i>Euglena</i> |
| (C) <i>Entamoeba</i> | (D) <i>Plasmodium</i> |

Q9. Silica shells are found on the surface of:

- (A) Marine Amoeboid Protozoans
- (B) Fresh water Amoeba
- (C) Parasitic Protozoans
- (D) Euglenoids

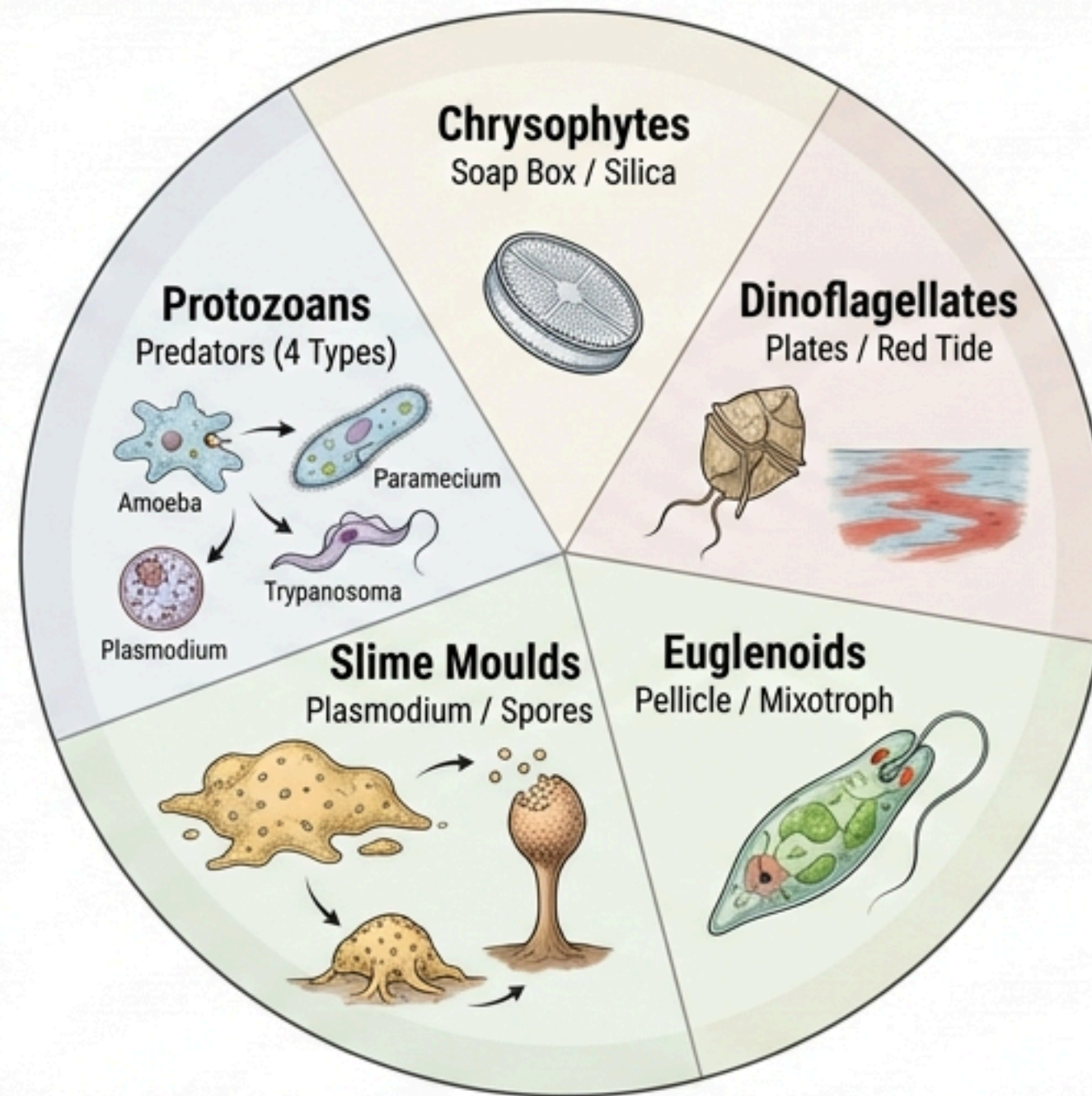
Q10. Which group forms a link with Plants, Animals, and Fungi?

- | | |
|------------|---------------------|
| (A) Monera | (B) Protista |
| (C) Fungi | (D) Archaeobacteria |

Answer Key & Explanations

- Q1 (B):** Incorrect because walls are embedded with **Silica**, not calcium carbonate.
- Q2 (B):** Statement II is false; plates are on the **outer** surface, not inner.
- Q3 (C):** Correct. (A is false: they have pellicle; B is false: pigments like **higher plants**).
- Q4 (B):** Both true, but R explains survival mechanism, not the *trigger* for fruiting bodies (which is unfavorable conditions).
- Q5 (B):** Both true. 'Chief producers' is due to photosynthetic capacity; Indestructibility is due to silica. They are independent facts.
- Q6 (B):** Active movement is due to cilia presence. The gullet is a feeding mechanism.
- Q7:** a-ii, b-iii, c-iv, d-i.
- Q8 (B):** *Euglena*.
- Q9 (A):** Marine amoeboid forms.
- Q10 (B):** Protista.

Chapter Summary: Kingdom Protista



Mastering Protista is the key to understanding the evolution of multicellular life.
Review the Comparison Table (Slide 10) for quick revision.